Properties of Laser Light (PLL)

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In this report, we examine many different properties of laser light. Firstly, the warm up behaviour for both unpolarised and polarised laser light will be examined, investigating the existence of longitudinal modes of oscillation in the cavity. Then we will look, qualitatively and quantitatively look at the spatial and temporal coherence of the laser light.

INTRODUCTION

Unfortunately, this is a short report so the lengthy history of the development of the laser cannot be fully treated however the key points will be covered. The theory of lasers was well understood way before its invention in 1951 by Charles H. Townes. Although there is heavy dispute around who invented the laser, and even a lawsuit around its patents, the first 'laser' was constructed in 1953, using a resonant microwave cavity. Nevertheless, the theory behind stimulated emission was well developed in the early 1920s along with the onset of quantum mechanics into the field of physics [1]. Using the old quantum theory framework, in a 1917 paper, Einstein theorised the existence of stimulated emission. He postulated that photons would prefer to travel together in the same state and therefore, spontaneously emitted photons, for which the name he gave the production of photons from electrons dropping down in energy levels, would then lead to or induce other photons to be emitted in the same direction (see Ref [2]).

Fast forward a decade, stimulated emission will be experimentally verified in 1928 by Rudolf Walther Ladenburg, who deemed it unpractical and he did not explore the subject any further [3]. The main subject of this report is the Helium Neon (He-Ne) laser which was invented by Javan, Benenett and Herriott in 1961 [4]. Surprisely, they did not win the Nobel Prize for their invention, rather the 1964 coveted prize went to the aforementioned Townes and his team for their foundational work on the maser and laser. Despite this, the He-Ne laser continues to be one of the most influential instruments in the modern development of technology.

The most common use of He-Ne lasers are in the field of optics. They are low cost and have comparatively easier modes of operation than their counter parts, producing beams of similar quality in terms of spatial coherence and coherence length [5]. A simple search on Google scholar for He-Ne laser will reveal a multitude of applications across various fields such as physics, biology and medicine.

THE HE-NE LASER

Quantum mechanics states that when an excited electron decays to a lower energy state, it will release a photon with energy equal to the energy difference of the two states. This is called spontaneous emission. When the emitted photon travels close to or pass by another excited electron, it may induce the same transition and the photon emitted from this interaction is the result of a stimulated emission process. The induced process will produce a photon with the exact same properties as the photon used to stimulate it and therefore, with the correct set up, a laser can emit one narrowband range of photons.

For lasers to operate, there must be a population inversion so that the rate of stimulated emission exceeds that of spontaneous emission or absorption. It follows that the number of electrons in an excited state will be greater than the number of electrons is a non-excited state. In a He-Ne laser, there is a mixture of neon and helium gas in a tube. The ratio between helium and neon atoms is 9:1. There is then an electrical discharge that ionises the gas, causing the helium atoms to enter metastable states. In this state, an electron in the helium atom is excited from 1s to 2s level. Because of quantum mechanics selection rules, this state is 'stable', despite it not being the lowest energy state. Therefore, the helium atoms are in a metastable state, where its downward transition is forbidden.

The photons emitted by the laser are generated with collisions between metastable helium atoms and neon atoms. Neon has 10 electrons residing in the 1s, 2s, 2p shells. The collision between a 2p neon electron and a metastable helium will excite the neon electron to higher energy states. The most likely state being 4s or 5s, rather than 3s or 3p. Once the collision occurs, the helium electron will fall back down to ground level. There are, consequently, many different wavelengths of light that can be possibly emitted from a tube of helium and neon gas, as seen in Figure 1

OPTICAL CAVITY

To select or filter out the desired wavelength, optical cavities are used in lasers. An optical cavity is two mirrors facing each other in a configuration that traps light. In the case of creating an operational laser, the light is trapped between a fully and a partially reflecting mirror, seen in Figure 2.

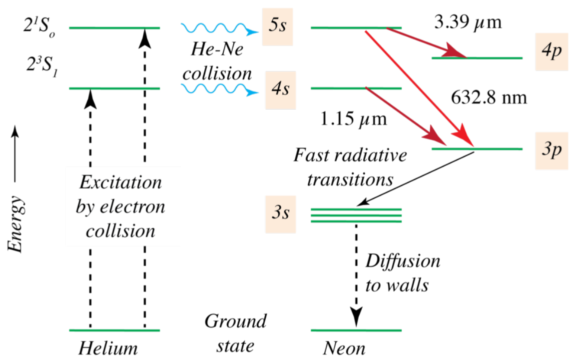

FIG. 1: Possible transitions in the interactions between an excited helium and neon atom. Source: Wikipedia [5]
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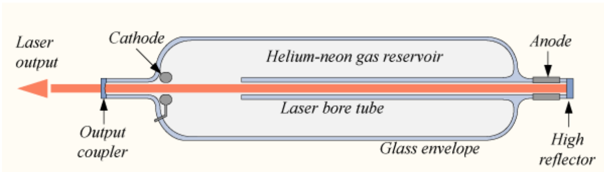


FIG. 2: Simple diagram showing the anatomy of the He-Ne laser. The cathode and anode provides the electrical discharge that excites the helium electrons inside the helium-neon gas reservoir that is contained by the glass envelope. The output coupler is a partially reflecting mirror of approximately 99% strength whilst the high reflector is approximately 99.99%. Therefore the laser output is 1% of the power of exhibited by the collisions. Source: RPMC lasers [6]

Longitudinal Modes of Oscillation

For a standing wave to exist, the path length of the light travelled between mirrors in the optical cavity must be an integer number of the wavelengths so $2L = m\lambda$. We can rewrite as a series of resonant frequencies

$$\nu = \frac{mc}{2L}. \tag{1}$$

The separation of these longitudinal modes is then

$$\Delta\nu = \frac{c}{2L}. \tag{2}$$

Coherence

Another property that laser lights possess is coherence. Coherence of light can be reduced to the simple fact: given a point in an optical field, $(\mathbf{r}_1, t_1)$, can one predict the value of another point in the same optical field $(\mathbf{r}_2, t_2)$ [7]. Because the spatial and temporal components of the field can be separated, we can analyse the coherence of the light in terms of temporal and spatial coherence.

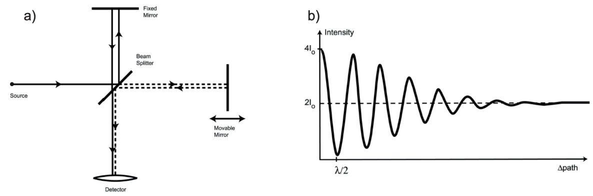


FIG. 3: (a) Schematic diagram of the Michelson interferometer setup. A source of light enters a beam splitter which then directs two beams into a fixed and a movable mirror. The two separate beams then are reflected back into the splitter where they recombine and enter the detector. (b) A graph displaying the intensity of the interferometry pattern based on the path difference, or the difference in distances between the two mirrors. Source: Peter Reece PHYS3114 Lectures

Temporal Coherence

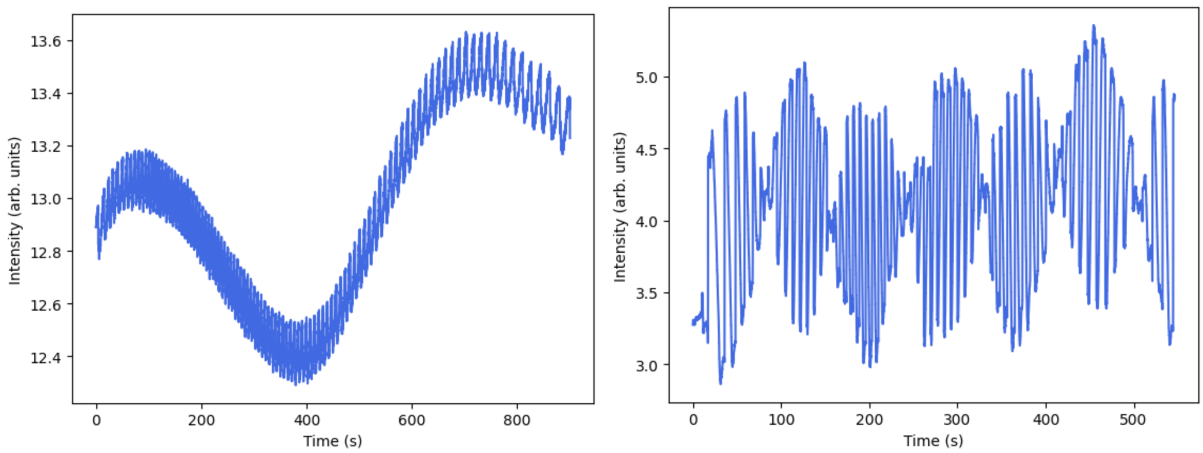
The most simple way to observe temporal coherence is in the Michelson interferometer. If we take a look at Figure 3, we can find the temporal coherence of light in effect. The visibility of interferometry fringes would oscillate but gradually converges as the path difference between the two beams from the same of light increases. In perfectly correlated and in phase waves, the intensity of the fringes would be twice the sum of the individual beams whereas in perfectly uncorrelated waves, the intensity of fringes will just be the sum of the two intensities.

We can hand wave a derivation for the visibility of fringes using the Wiener-Khinchin theorem which states an autocorrelation (coherence function) is the Fourier transform of the power spectrum. If we take the power spectrum to be a series of Dirac Delta functions with a spacing of $\Delta\nu$, where ν is the longitudinal mode spacing. We will find the magnitude of the coherence function (proportional to the visibility) to have a periodic factor of $\tau_{\text{rev}} = \frac{1}{\Delta\nu}$. Therefore, the coherence length can be written in terms of the cavity length,

$$\Delta_{\text{rev}} = c\tau_{\text{rev}} = \frac{c}{\Delta\nu} = 2L_{\text{cav}}. \tag{3}$$

Spatial Coherence

Spatial coherence is essentially the measure of coherence in the transverse direction, for two points on the plane of the light's oscillation. This is a simple construction as we can just investigation this coherence by checking the interference pattern at these two points using the Young double slit experiment. Using the same result as the temporal coherence, if the two light waves are correlated their intensity sum will be much higher (twice) than the uncorrelated light waves.



(a) Without polarisation

(b) With polarisation

FIG. 4: Experimental measurement for the intensity of the laser while it is warming up. There are two plots, one where there is no polariser and one where the polarisation is changed by 10° every 10 seconds for 9 minutes. In the plot without polarisation, we observe a slowly drifting oscillating wave. In the plot with polarisation, there is a modulated wave.

EXPERIMENTAL RESULTS & ANALYSIS

Measurement of the wavelength of the laser

The wavelength of the laser can be calculated using the Young double slit experiment. Placing a grating with 1.59μm spacing in front of the laser will cause a first order maximum at 15 cm. The distance from the grating to the screen is 40 cm and so using

$$\lambda = d \sin\theta \approx \frac{dy}{L}. \tag{4}$$

We obtain a value for the wavelength of the laser to be 633.75 $\pm$0.5 nm. This does not agree with the given quantity of 632.8 nm, representing a 0.15% error.

Laser warm up

We wanted to first test the findings of Ref [8]. They observed then when He-Ne lasers are approaching 100% of the total usable output, there is fluctuations in the intensity measured. This is due to the changes in the length of the cavity due to thermal expansion. This thermal expansion would cause the longitudinal modes to drift across the Doppler gain curve and each time the peak would change modes, the intensity of the light will oscillate, producing the warm up beat seen in 4.

In the plot of the warm up with a rotating polariser we find the same oscillations as before however now with a modulating wave on top. This modulating wave comes from the slowly varying amplitude of the faster beat frequency carrier wave which we cannot directly see here. The slowly drifting thermal expansion can still be seen as the wavepackets are also up and down in amplitude. The beat frequency comes from the difference in the two frequencies of the two orthogonal polarisations of light originating in the cavity. Therefore, we can only see the summation of these two axes of polarisation using a linear polariser. This beat frequency is further explored in the next section.

Longitudinal modes

We can observe this beat frequency more closely by rotating the polariser whilst measuring the laser output with a spectrum analyser. We find two peaks within the range of the spectrum analyser, one at 687 MHz and the other at 1374 MHz. This corresponds to the beat pairs between ν_0 and ν_1 and the next pair ν_0 and ν_2 with the spacing between modes corresponding to Equation 2. We can calculate the length of the cavity using the beat frequency,

$$L = \frac{3 \times 10^8}{2 \times 687.183 \times 10^6} = 21.8 \text{ cm}. \tag{5}$$

If we rotate the polariser, we can observe the behaviour of the two orthogonal polarisers. The polariser begins where the peak is at a minimum. The minimum occurs as the interference between the two orthogonal modes is zero. We then find max peaks to occur at a polarisation rotation angle of 39°, 146°, 220° and 317° and minimum peaks to occur at 0°, 85°, 180° and 267°. We can roughly see the mixing of the two orthogonal modes here. At points where one mode is favoured, i.e $\approx$ 0°, 90°, 180°, 270°, we observe minimums as the interference and hence beat frequency is 0. At angles where the orthogonal modes can be evenly mixed, i.e $\approx$ 45°, 135°, 225° and 315°, the beat frequency is at a maximum. The experimental results roughly agree with this range.

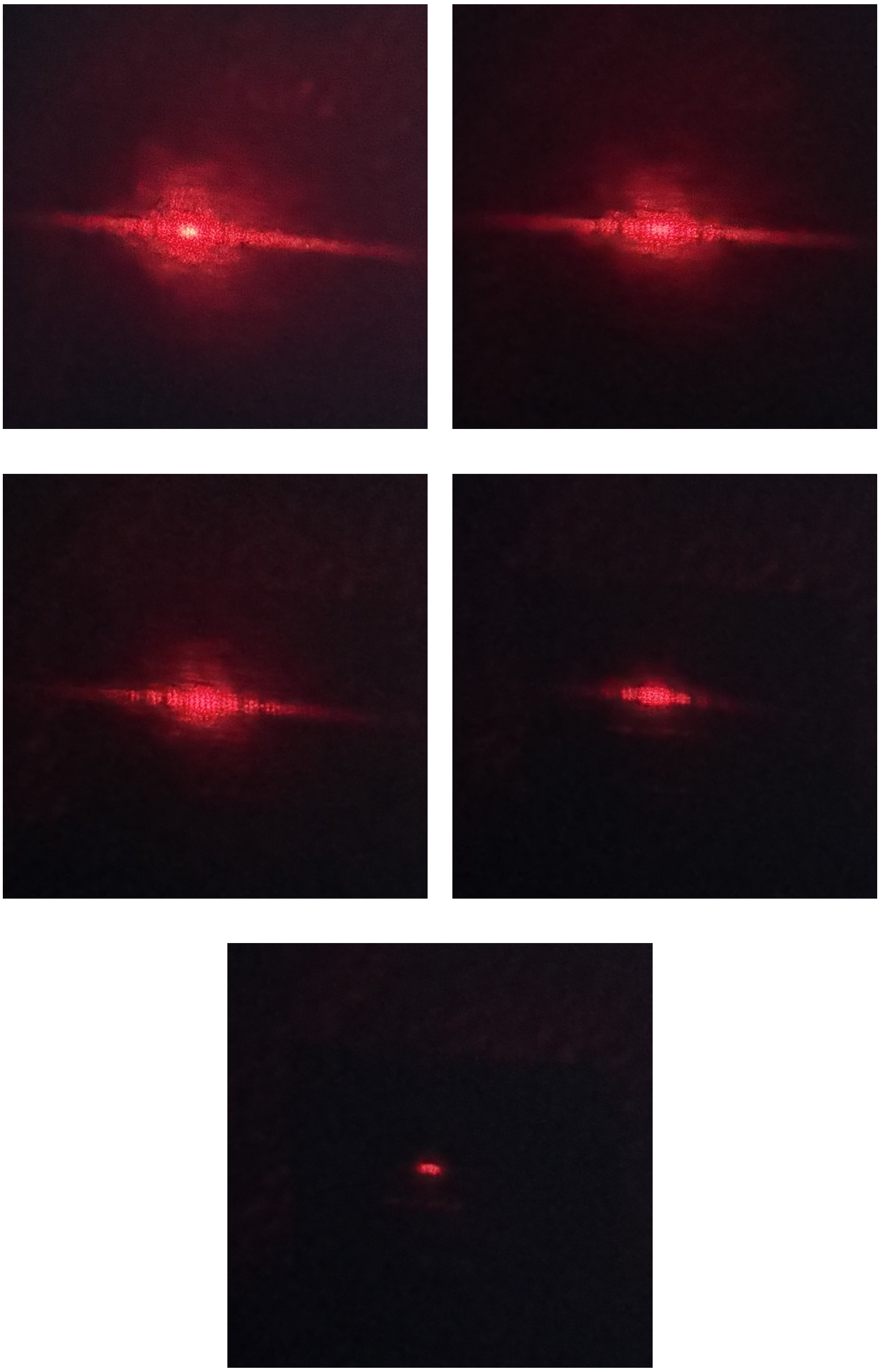


FIG. 5: From left to right and top to bottom sequence, the grating is translated towards one direction slowly, increment not noted. The interference fringes disappears and the total intensity of the pattern falls.

Spatial coherence

A qualitative analysis of spatial coherence is given here. We can observe the decrease in intensity with a shift in the transverse direction of the grating in Figure 5. As we scan the grating along the transverse plane of the laser, the intensity of the laser falls, indicating the decrease in the interference between the points on the plane.

Temporal Coherence

The temporal coherence is measured in Figure 6. We can observe an oscillation in visibility as the path difference increases. The amplitude of the intensity should decrease and it can sort of be observed in the figure however there is a big spike near the 10 cm point, this can be attributed to a gentle bump in the set up.

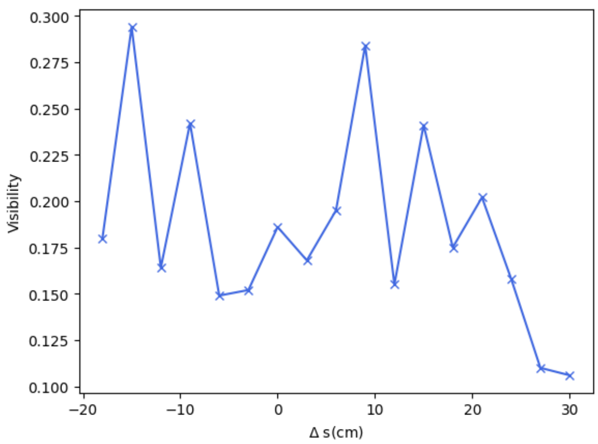


FIG. 6: Experimental graph of the visibility versus change in mirrors difference in path length, same plot as in the theoretical figure 3.

The coherence length can be calculated by using the distance from the first peak to the first zero. From our figure, the max peak roughly occurs at path difference of -15 cm and the first zero at +30 cm. The coherence length is then 45 cm. The cavity using this method and Equation 3 is measured to be 22.5 cm. Therefore this is a 3% disagreement between the two methods for the length of the cavity.

CONCLUSION

In this short report, we were able to examine multiple properties of laser light including measurement of the wavelength, interactions between polarisation and longitudinal modes as well as coherence. We found the cavity length using two different methods and found a 3% disagreement between the two, verifying the theoretical framework to a strong degree.

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